

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #3

Instructions

- TA in charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (25%)	Question 2 (25%)	Question 3 (20%)	Question 4 (15%)	Question 5 (15%)	Total

Question 1 (25%)

Design a minimum-cost system that performs the addition (S) of two 4-bit unsigned numbers: $A = \{a_3a_2a_1a_0\}$ and $B = \{b_3b_2b_1b_0\}$ as follows. If the input number is odd, it will be included in the addition, alternatively, if the number is even, it will not be included in the addition. Use only HAs, FAs, and the constants '0' and '1'.

Question 2 (25%)

Design a minimal-cost ternary (base 3) MUX (TMUX). The TMUX receives as input three 1-digit ternary numbers each represented by binary signals: $t_0 = \{a_1a_0\}$, $t_1 = \{b_1b_0\}$, $t_2 = \{c_1c_0\}$, and a 1-digit selector $t_s = \{s_1s_0\}$. The TMUX produces a 1-digit ternary output $t_z = \{z_1z_0\}$.

Question 3 (20%)

(a) Design and implement the following system:

Inputs: $X = x_3x_2x_1x_0$ and $S = s_1s_0$

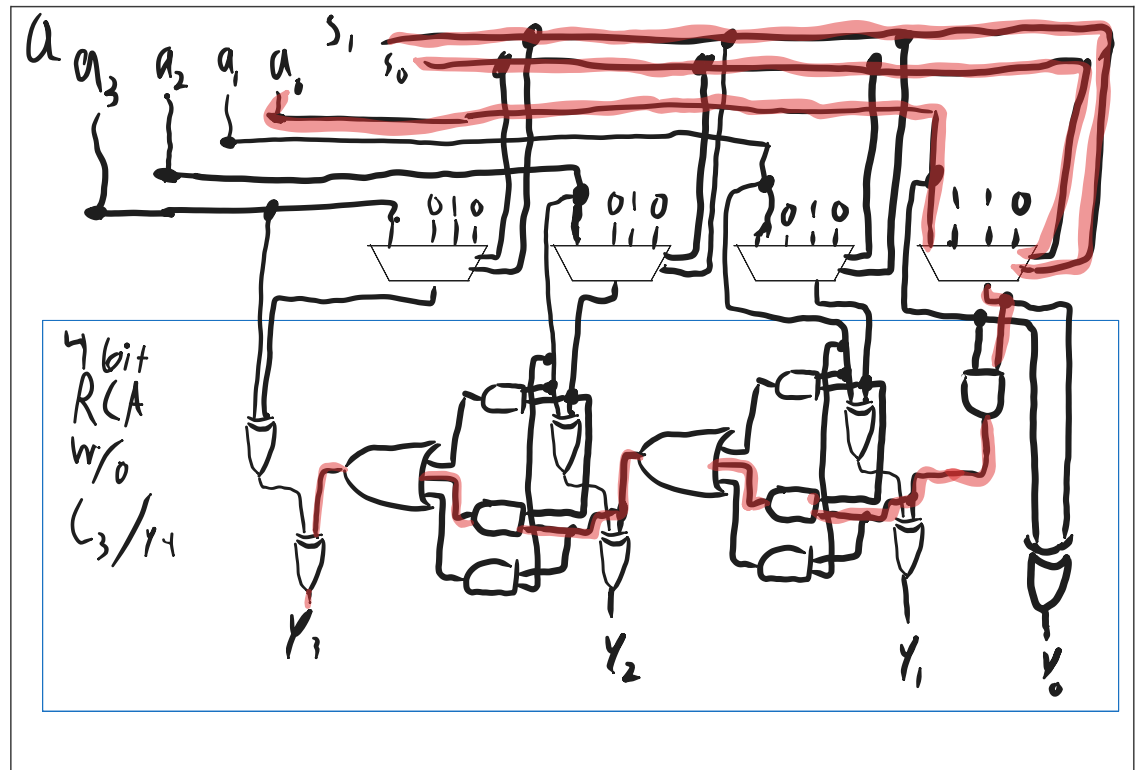
Output: $Y = y_3y_2y_1y_0$

The functionality of the system is described in the table below. You can ignore any overflow that may occur. You can only use primitive gates, full-adders, half-adders, and 2-1 and 4-1 multiplexers. Any multiplexer can be considered a single gate, no need to implement. Alternatively, provide the internal circuitry of the half- and full-adders, these cannot be considered as a single gate. Hint: To implement a multiplication by two, consider shifting the output bits.

s_1s_0	Action
00	$Y = X$

00	$Y = X$
01	$Y = X - 1$
10	$Y = X + 1$
11	$Y = 2 \cdot X$ (multiply by two)

(b) Mark the critical path on the circuit you designed. What is the delay of the critical path? Assume that delay of a path is defined as the number of gates on the path. Also, assume that a multiplexer has the same delay as other primitive gates (this assumption is not true in most realistic cases and is only provided here to simplify the analysis).



b Critical path marked in red
 passes through 7 gates, 1 4-1 mux,
 3 and
 2 or
 1 xor

Question 4 (15%)

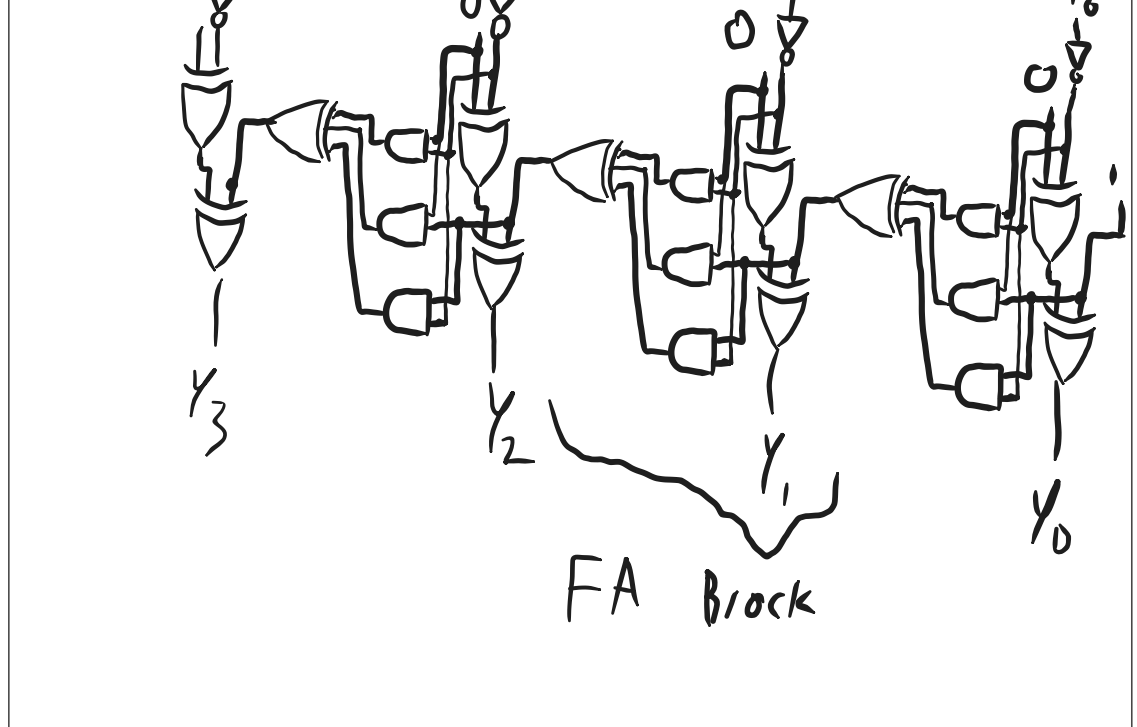
Design and implement a circuit that generates the (15)'s complement of a HEX digit $p(p_3p_2p_1p_0)$. Provide all of the design steps.

$$K_{15} = (16 - 1) - p$$

$$= (F) - p$$

do subtraction in 2's complement
 using RCA with inverse of $p(1000) - (p_3p_2p_1p_0)$
 & $c_{in} = 1$ to handle 2's
 conversion comp

$$\begin{array}{cccc}
 & p_3 & & p_2 \\
 1 & \downarrow & & \downarrow \\
 & & & & p_1 & & p_0
 \end{array}$$



Question 5 (15%)

Suppose that we want to determine how many of the bits in an eight-bit unsigned number are equal to 1. Design the minimum-cost circuit that can accomplish this task using only half- and full-adders.

